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Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 3– Réseaux II

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

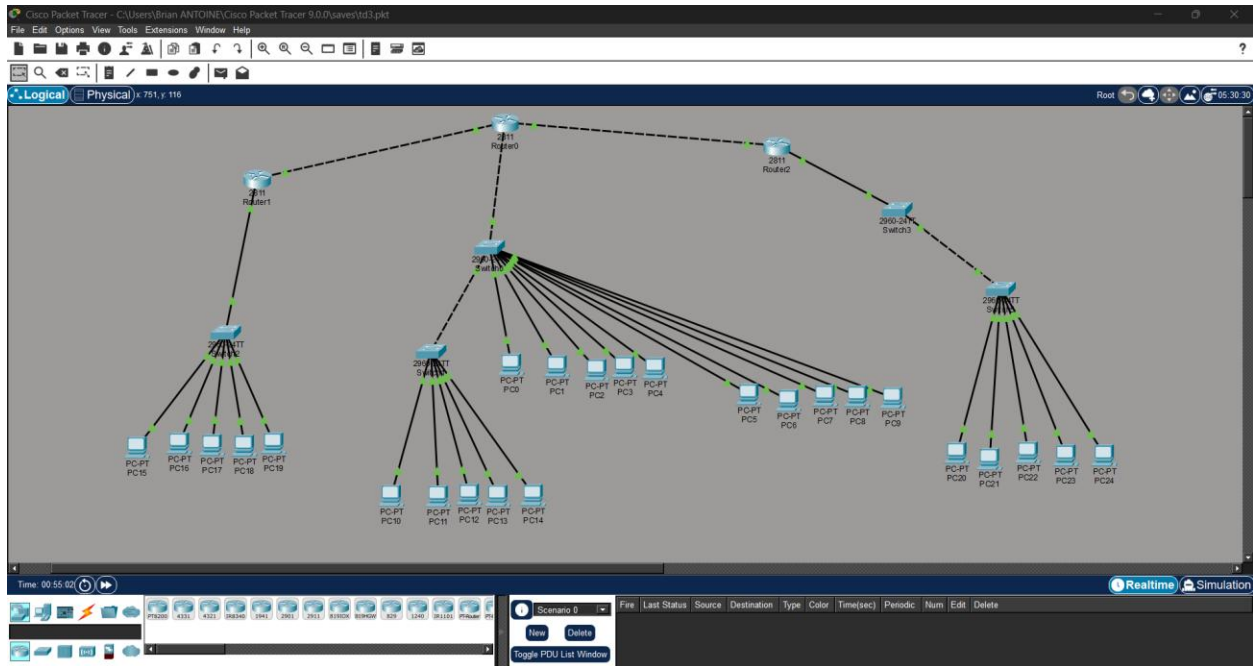
Niveau : L3

Date: 03/04/2026

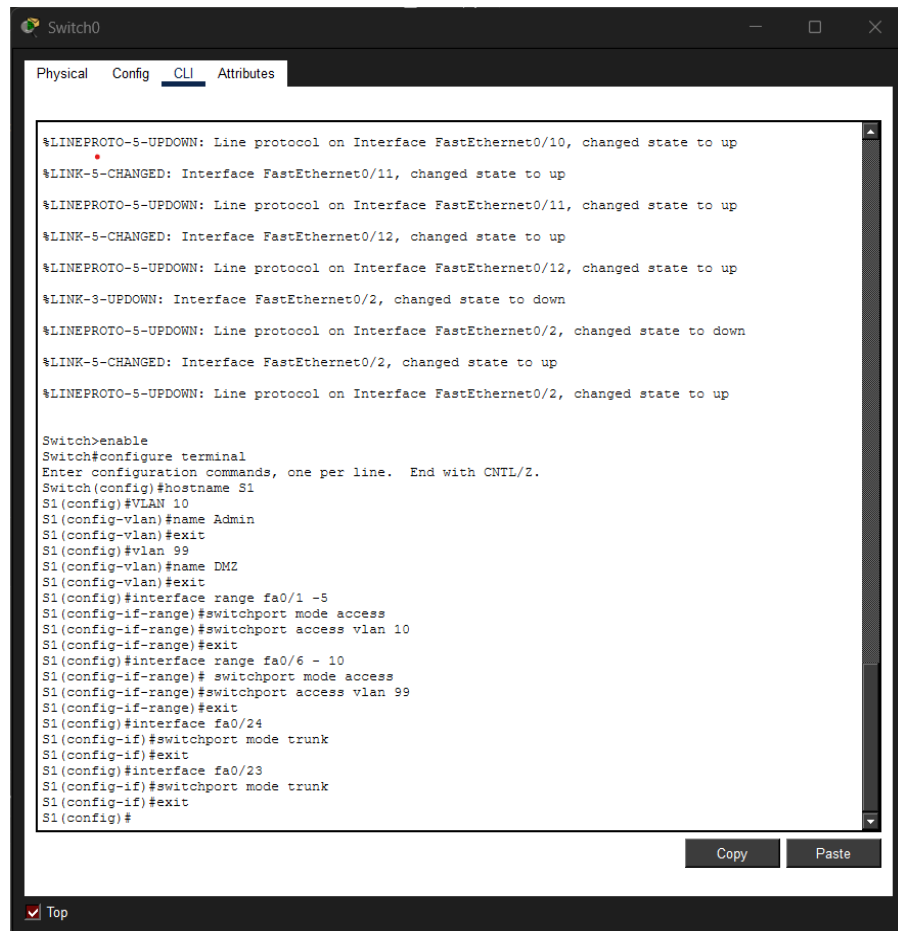
L'objectif de ce TD est de :

1. Apprendre à créer et gérer plusieurs VLANs pour isoler les différents départements (Administration, Commercial, Technique, Invités, DMZ).
2. Comprendre le routage inter-VLAN à l'aide de routeurs et de sous-interfaces.
3. Identifier les avantages de la segmentation pour la sécurité.
4. Configurer des mots de passe forts pour l'accès console et VTY.
5. Activer SSH sur les routeurs pour un accès sécurisé à distance.
6. Appliquer la port-security sur les switchs pour limiter les adresses MAC autorisées sur chaque port.
7. Créer des ACL étendues pour contrôler l'accès entre VLANs (ex. : VLAN invités ne peuvent pas accéder à la DMZ ou aux ressources sensibles).
8. Filtrer le trafic réseau de façon sélective (protocole, IP source/destination).
9. Simulation de scénarios d'attaque et tests

1. Creation de la Topologie : 3 routeurs, 5 switches et 25 pcs



2. Configuration des Vlans sur les switchs

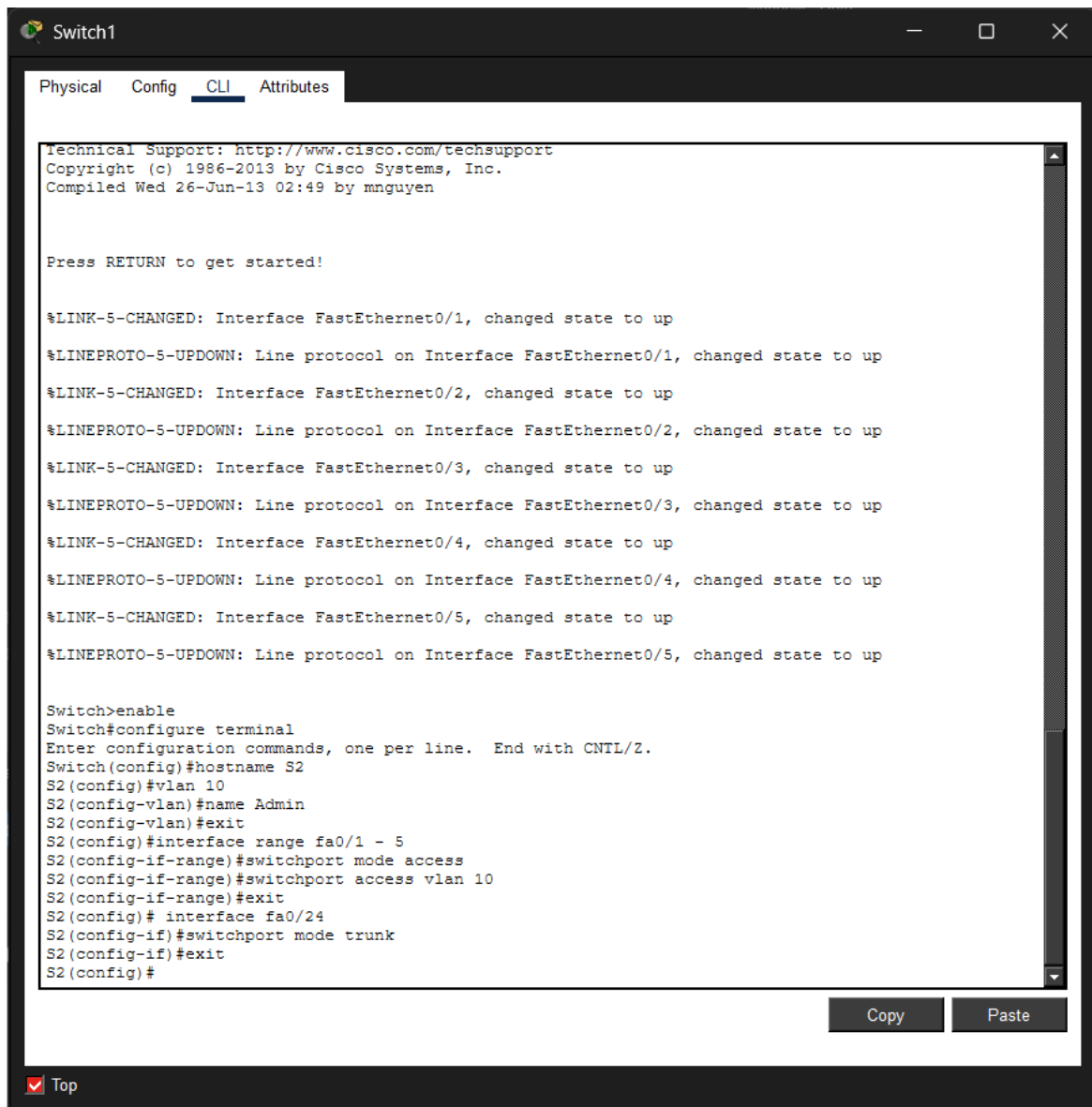


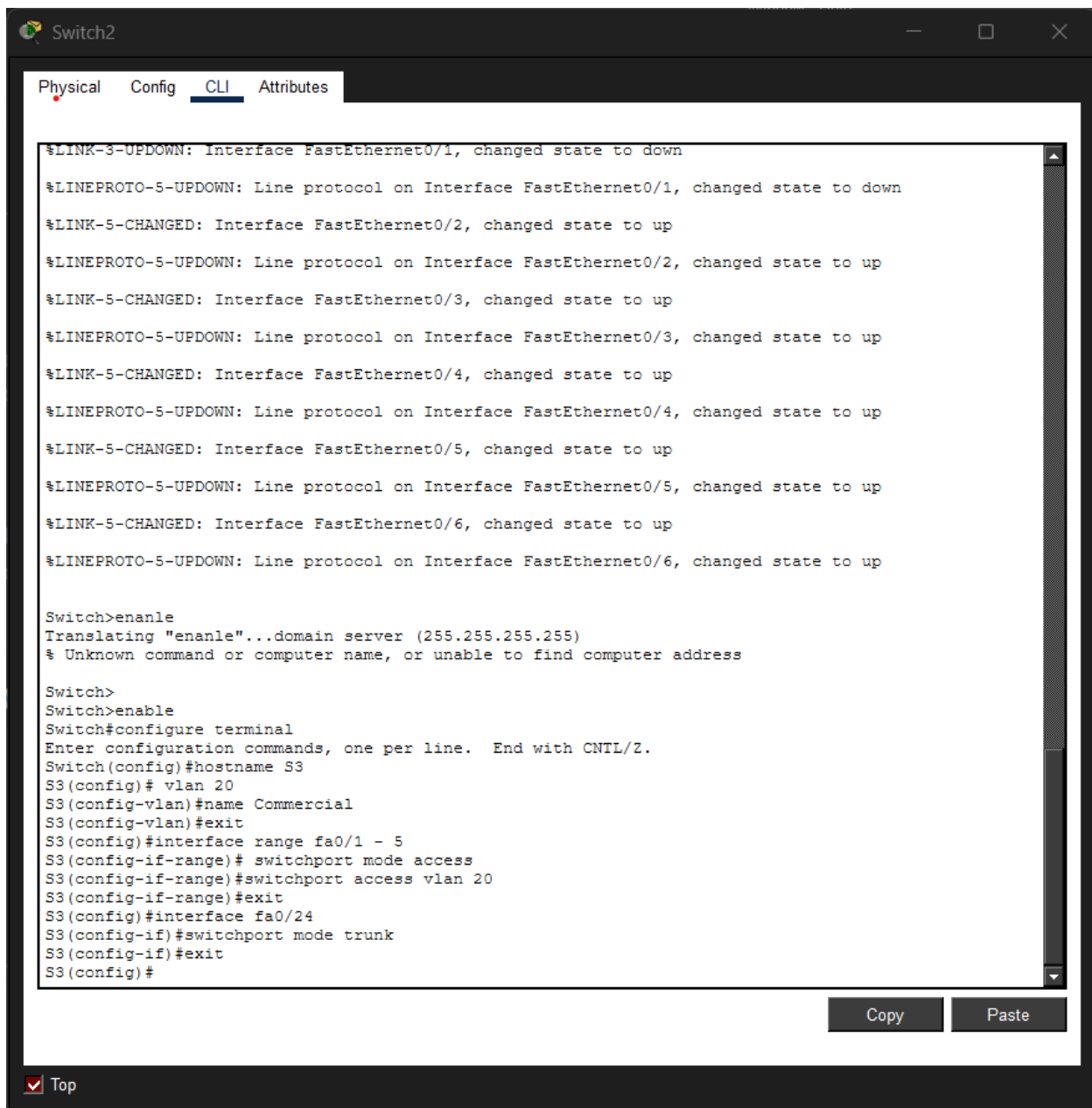
The screenshot shows a network switch configuration window titled "Switch0". It has four tabs: "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of status messages and configuration commands. The status messages at the top indicate changes in line protocol and link states for various interfaces. Below these, the configuration commands are listed, starting with "enable" and "configure terminal", followed by setting the hostname to "S1", creating VLANs 10 and 99, and configuring interfaces fa0/1-5, fa0/6-10, fa0/24, and fa0/23.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#VLAN 10
S1(config-vlan)#name Admin
S1(config-vlan)#exit
S1(config)#vlan 99
S1(config-vlan)#name DMZ
S1(config-vlan)#exit
S1(config)#interface range fa0/1 -5
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 10
S1(config-if-range)#exit
S1(config)#interface range fa0/6 - 10
S1(config-if-range)# switchport mode access
S1(config-if-range)#switchport access vlan 99
S1(config-if-range)#exit
S1(config)#interface fa0/24
S1(config-if)#switchport mode trunk
S1(config-if)#exit
S1(config)#interface fa0/23
S1(config-if)#switchport mode trunk
S1(config-if)#exit
S1(config)#
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button with a checkmark icon.





Switch3

PhysicalConfigCLIAttributes

Switch	Ports	Model	SW Version	SW Image
*	1 26	WS-C2960-24TT-L	15.0(2)SE4	C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

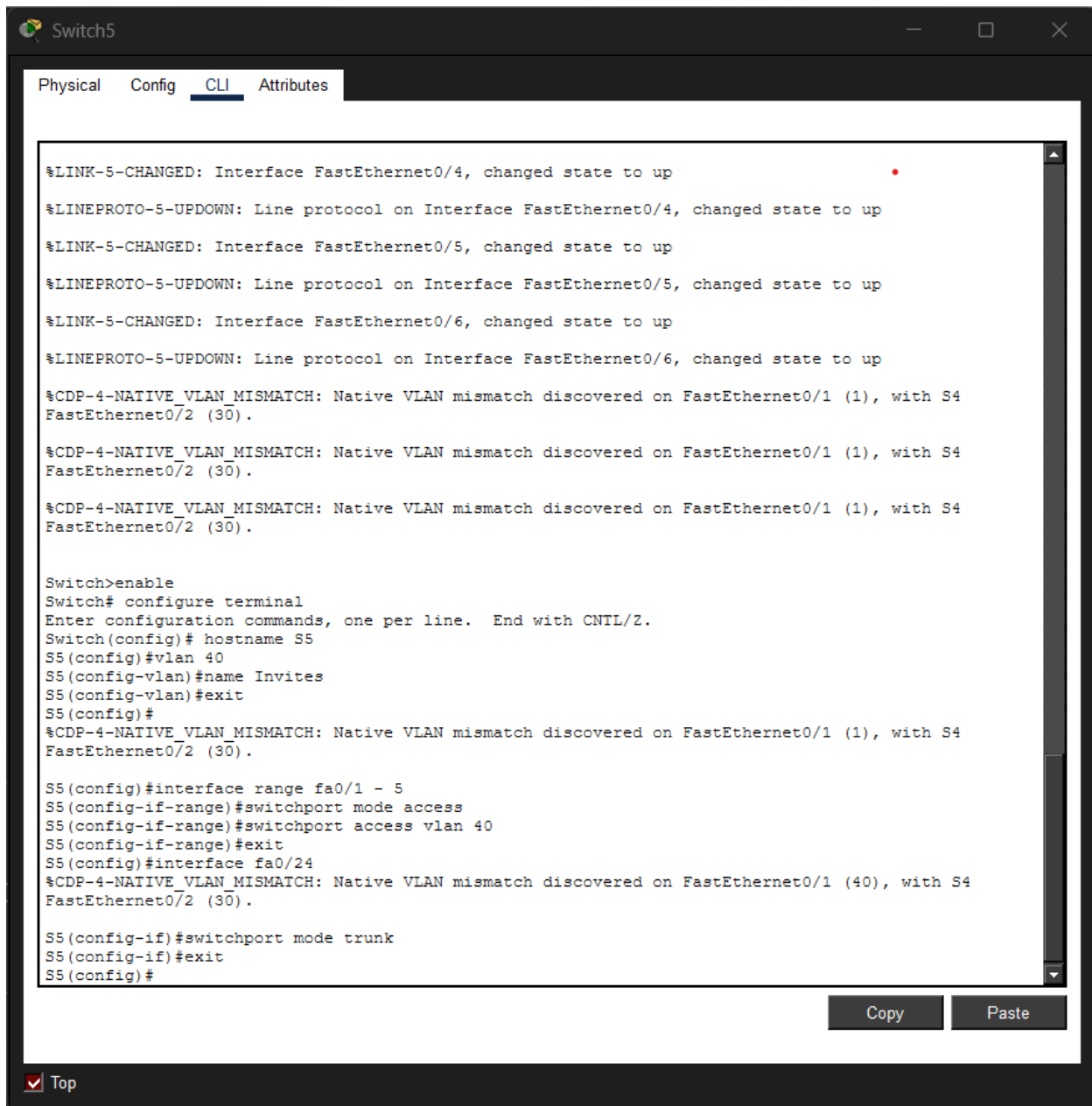
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S4
S4(config)#vlan 30
S4(config-vlan)#name Technique
S4(config-vlan)#exit
S4(config)#interface range fa0/1 - 5
S4(config-if-range)#switchport mode access
S4(config-if-range)#switchport access vlan 30
S4(config-if-range)#exit
S4(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with Switch
FastEthernet0/1 (1).
interface fa0/23
S4(config-if)#interface fa0/23
S4(config-if)#switchport mode trunk
S4(config-if)#exit
S4(config)#interface fa0/24
S4(config-if)# switchport mode trunk
S4(config-if)#exit
S4(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with Switch
FastEthernet0/1 (1).
```

CopyPaste

Top



The screenshot shows a network switch CLI window titled "Switch5". The "CLI" tab is selected, displaying a series of status messages and configuration commands. The status messages indicate that interfaces FastEthernet0/4, 0/5, and 0/6 have changed state to up, and line protocols on these interfaces have also changed state to up. There are also three CDP-4-NATIVE_VLAN_MISMATCH messages discovered on FastEthernet0/1 (1), with S4 FastEthernet0/2 (30).

```
Switch>enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname S5
S5(config)#vlan 40
S5(config-vlan)#name Invites
S5(config-vlan)#exit
S5(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with S4
FastEthernet0/2 (30).

S5(config)#interface range fa0/1 - 5
S5(config-if-range)#switchport mode access
S5(config-if-range)#switchport access vlan 40
S5(config-if-range)#exit
S5(config)#interface fa0/24
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (40), with S4
FastEthernet0/2 (30).

S5(config-if)#switchport mode trunk
S5(config-if)#exit
S5(config)#
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button with a red checkmark icon.

3. Configuration des routeurs



Router0

Physical Config CLI Attributes

If you require further assistance please contact us by sending email to export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MT2 (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

.

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface fa0/0
R1(config-if)# no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#interface fa0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface fa0/0.99
R1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.99, changed state to down

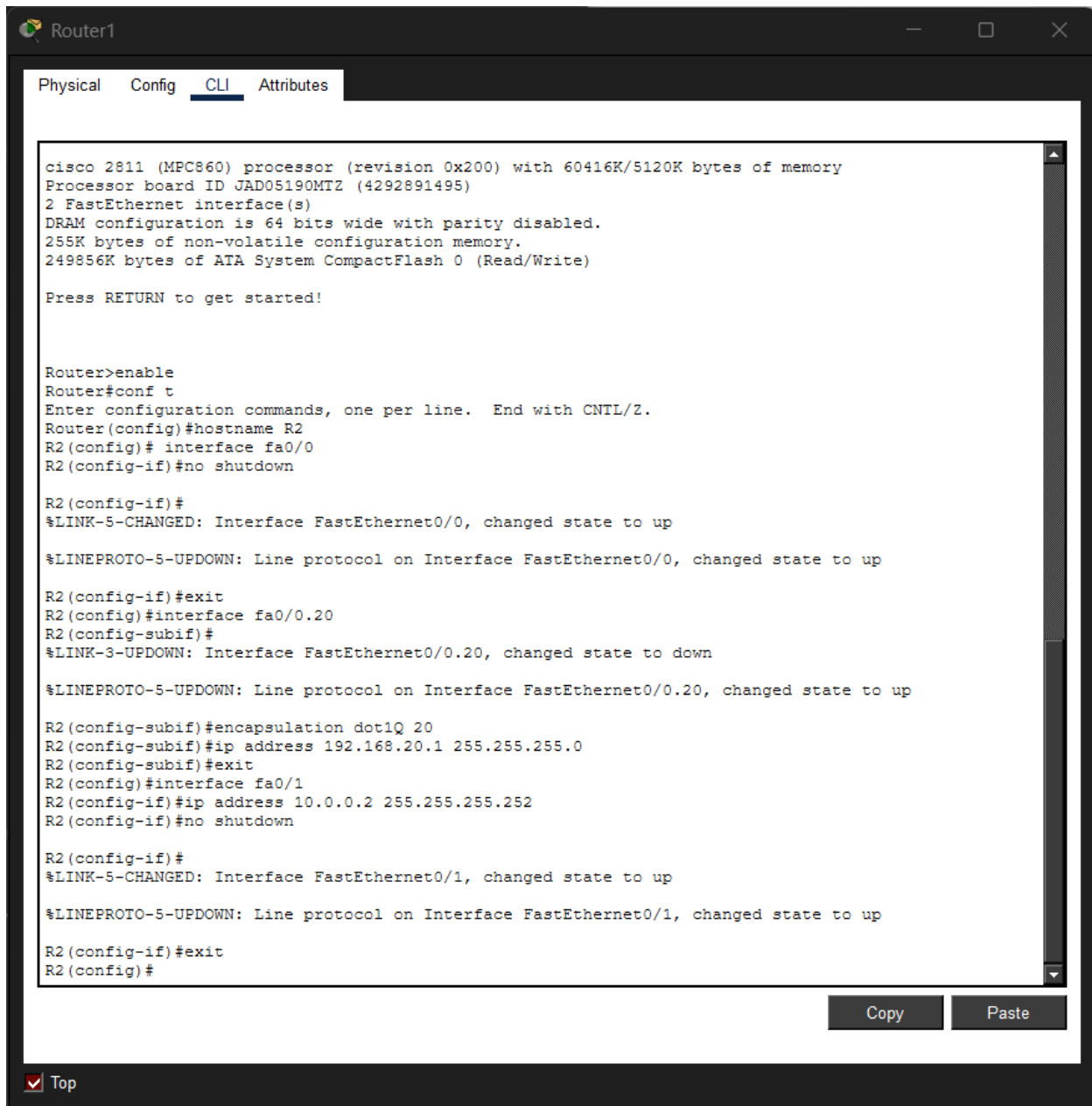
R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.99.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface fa0/1
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown

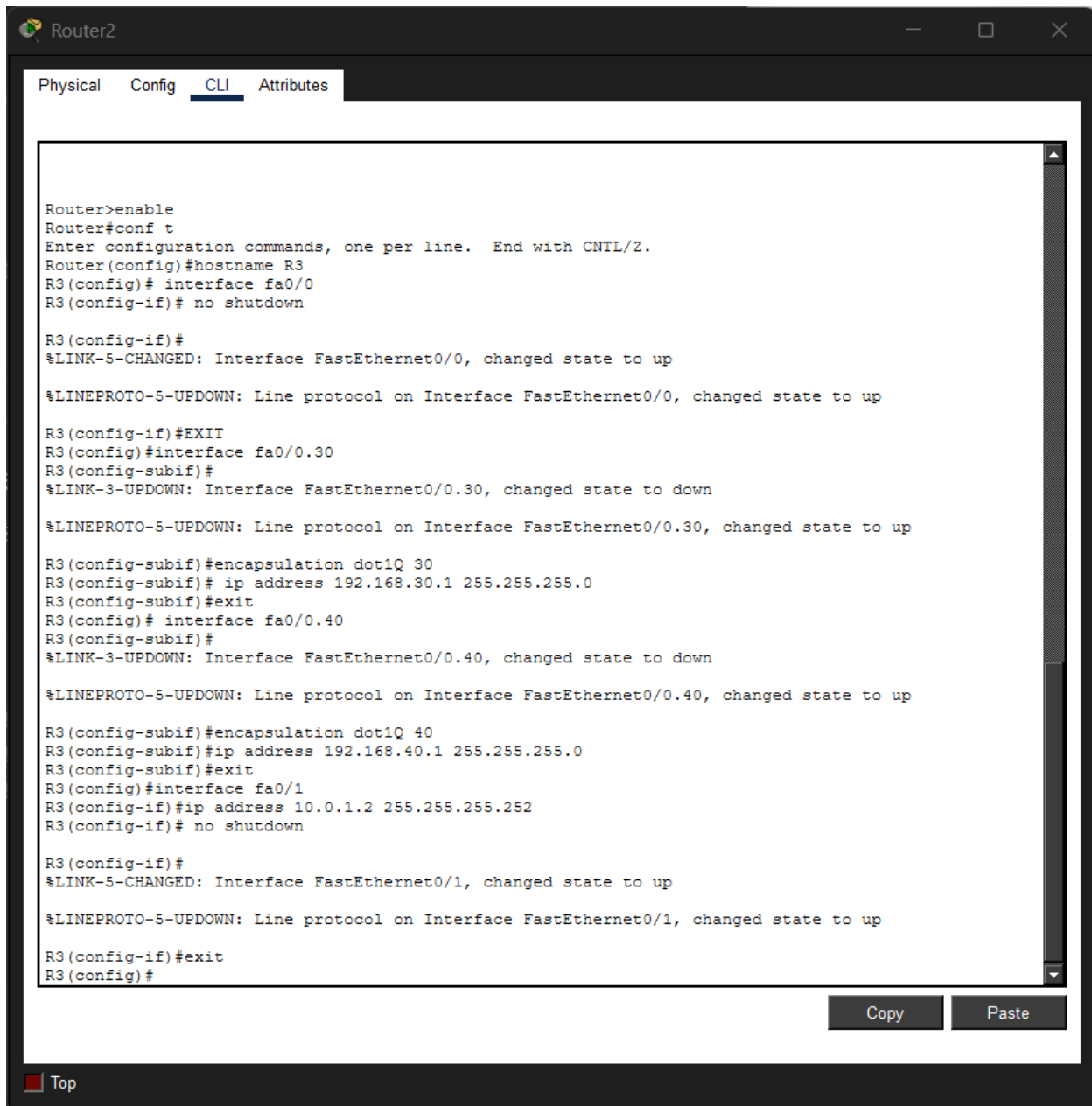
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

Copy

Paste

☒ Top





The screenshot shows a terminal window titled "Router2" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of configuration commands and their outputs. The commands include enabling the router, entering configuration mode, setting the hostname to R3, configuring interface fa0/0, and then configuring subinterfaces fa0/0.30, fa0/0.40, and fa0/1. Each subinterface is configured with a specific encapsulation, IP address, and no shutdown. The output shows the state changes for each interface, such as "Interface FastEthernet0/0, changed state to up" and "Line protocol on Interface FastEthernet0/0.30, changed state to up". At the bottom right, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)# interface fa0/0
R3(config-if)# no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R3(config-if)#EXIT
R3(config)#interface fa0/0.30
R3(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up

R3(config-subif)#encapsulation dot1Q 30
R3(config-subif)# ip address 192.168.30.1 255.255.255.0
R3(config-subif)#exit
R3(config)# interface fa0/0.40
R3(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.40, changed state to up

R3(config-subif)#encapsulation dot1Q 40
R3(config-subif)#ip address 192.168.40.1 255.255.255.0
R3(config-subif)#exit
R3(config)#interface fa0/1
R3(config-if)#ip address 10.0.1.2 255.255.255.252
R3(config-if)# no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R3(config-if)#exit
R3(config)#
```

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➤ Vérification du routage

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, FastEthernet0/1
L       10.0.0.1/32 is directly connected, FastEthernet0/1
C       10.0.1.0/30 is directly connected, FastEthernet0/0
L       10.0.1.1/32 is directly connected, FastEthernet0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0.10
L       192.168.10.1/32 is directly connected, FastEthernet0/0.10
S       192.168.20.0/24 [1/0] via 10.0.0.2
S       192.168.30.0/24 [1/0] via 10.0.1.2
S       192.168.40.0/24 [1/0] via 10.0.1.2
    192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
--More--
```

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Router0

Physical Config **CLI** Attributes

```
% Ambiguous command: "p route 192.168.30.0 255.255.255.0 10.0.1.2"
R1(config)#ip route 192.168.30.0 255.255.255.0 10.0.1.2
R1(config)#ip route 192.168.40.0 255.255.255.0 10.0.1.2
R1(config)#show ip route
R1(config)#
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

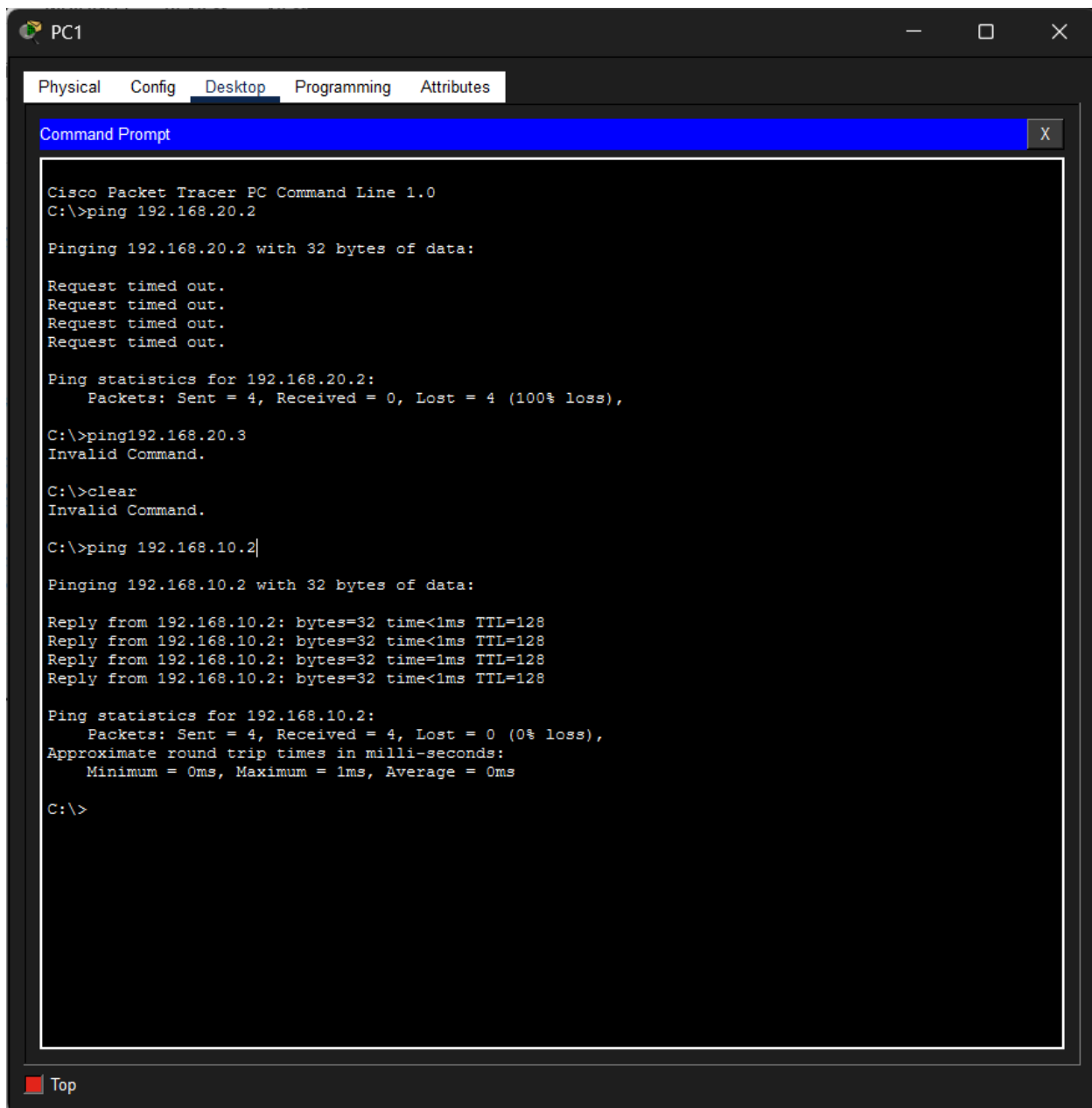
    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, FastEthernet0/1
L       10.0.0.1/32 is directly connected, FastEthernet0/1
C       10.0.1.0/30 is directly connected, FastEthernet0/0
L       10.0.1.1/32 is directly connected, FastEthernet0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0.10
L       192.168.10.1/32 is directly connected, FastEthernet0/0.10
S       192.168.20.0/24 [1/0] via 10.0.0.2
S       192.168.30.0/24 [1/0] via 10.0.1.2
S       192.168.40.0/24 [1/0] via 10.0.1.2
    192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.99.0/24 is directly connected, FastEthernet0/0.99
L       192.168.99.1/32 is directly connected, FastEthernet0/0.99

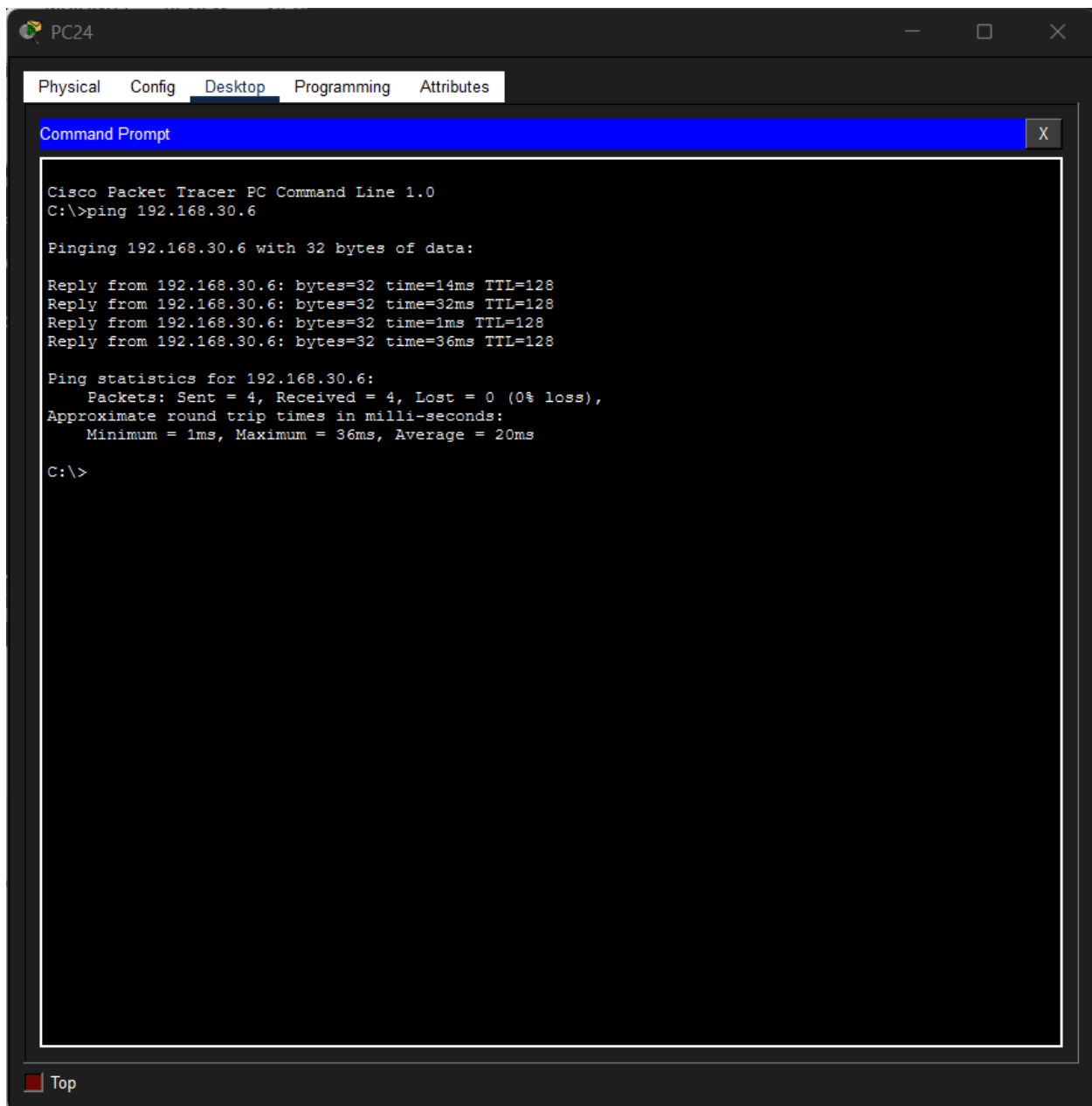
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0  10.0.1.1        YES manual up          up
FastEthernet0/0.10 192.168.10.1    YES manual up          up
FastEthernet0/0.99 192.168.99.1    YES manual up          up
FastEthernet0/1   10.0.0.1        YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
R1#
```

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Top

➤ Tests sur les PC : ping ip





- **Bloquer les invités (VLAN 40)** d'accéder à la DMZ (VLAN 99) et à l'Admin (VLAN 10).

Router2

Physical Config CLI Attributes

```
R3(config-if)#exit
R3(config)#ip route 192.168.10.0 255.255.255.0 10.0.1.1
R3(config)#ip route 192.168.99.0 255.255.255.0 10.0.1.1
R3(config)# ip route 192.168.20.0 255.255.255.0 10.0.1.1
R3(config)#

R3 con0 is now available

Press RETURN to get started.

R3>enable
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 120 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
R3(config)#access-list 120 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
R3(config)#access-list 120 permit ip any any
R3(config)#interface fa0/0.40
R3(config-subif)#ip access-group 120 in
R3(config-subif)#exit
R3(config)#
```

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```
R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
R1(config)#access-list 110 permit ip any any
R1(config)# interface fa0/0.99
R1(config-subif)#ip access-group 110 in
R1(config-subif)#exit
R1(config)#
```

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Port security sur les switches

- ✓ Limite à **2 appareils maximum** par port, avec apprentissage automatique des adresses MAC.

```
S1>enable
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface range fa0/1 - 22
S1(config-if-range)#switchport port-security
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
S1(config-if-range)#switchport port-security maximum 2
S1(config-if-range)#switchport port-security violation restrict
S1(config-if-range)#switchport port-security mac-address sticky
S1(config-if-range)#exit
S1(config)#
```

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Top

```
S2>enable
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface range fa0/1 - 22
S2(config-if-range)#switchport port-security
Command rejected: FastEthernet0/6 is a dynamic port.
Command rejected: FastEthernet0/7 is a dynamic port.
Command rejected: FastEthernet0/8 is a dynamic port.
Command rejected: FastEthernet0/9 is a dynamic port.
Command rejected: FastEthernet0/10 is a dynamic port.
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
S2(config-if-range)#switchport port-security maximum 2
S2(config-if-range)#switchport port-security violation restrict
S2(config-if-range)#switchport port-security mac-address sticky
S2(config-if-range)#exit
S2(config)#
```

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Top

```
S4>enable
S4#conf t
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with S5
FastEthernet0/1 (40).

Enter configuration commands, one per line. End with CNTL/Z.
S4(config)#interface range fa0/1 - 22
S4(config-if-range)#switchport port-security
Command rejected: FastEthernet0/6 is a dynamic port.
Command rejected: FastEthernet0/7 is a dynamic port.
Command rejected: FastEthernet0/8 is a dynamic port.
Command rejected: FastEthernet0/9 is a dynamic port.
Command rejected: FastEthernet0/10 is a dynamic port.
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
S4(config-if-range)#switchport port-security maximum 2
S4(config-if-range)#switchport port-security violation restrict
S4(config-if-range)#switchport port-security mac-address sticky
S4(config-if-range)#exit
S4(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (30), with S5
FastEthernet0/1 (40).
```

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```
S5>enable
S5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S5(config)# interface range fa0/1 - 22
S5(config-if-range)#switchport port-security
Command rejected: FastEthernet0/6 is a dynamic port.
Command rejected: FastEthernet0/7 is a dynamic port.
Command rejected: FastEthernet0/8 is a dynamic port.
Command rejected: FastEthernet0/9 is a dynamic port.
Command rejected: FastEthernet0/10 is a dynamic port.
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
S5(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (40), with S4
FastEthernet0/2 (30).
switchport port-security maximum 2
S5(config-if-range)# switchport port-security violation restric
S5(config-if-range)# switchport port-security mac-address sticky
S5(config-if-range)#exit
S5(config)#
```

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```

S3>enable
S3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#interface range fa0/1 - 22
S3(config-if-range)#switchport port-security
Command rejected: FastEthernet0/6 is a dynamic port.
Command rejected: FastEthernet0/7 is a dynamic port.
Command rejected: FastEthernet0/8 is a dynamic port.
Command rejected: FastEthernet0/9 is a dynamic port.
Command rejected: FastEthernet0/10 is a dynamic port.
Command rejected: FastEthernet0/11 is a dynamic port.
Command rejected: FastEthernet0/12 is a dynamic port.
Command rejected: FastEthernet0/13 is a dynamic port.
Command rejected: FastEthernet0/14 is a dynamic port.
Command rejected: FastEthernet0/15 is a dynamic port.
Command rejected: FastEthernet0/16 is a dynamic port.
Command rejected: FastEthernet0/17 is a dynamic port.
Command rejected: FastEthernet0/18 is a dynamic port.
Command rejected: FastEthernet0/19 is a dynamic port.
Command rejected: FastEthernet0/20 is a dynamic port.
Command rejected: FastEthernet0/21 is a dynamic port.
Command rejected: FastEthernet0/22 is a dynamic port.
S3(config-if-range)#switchport port-security maximum 2
S3(config-if-range)#switchport port-security violation restrict
S3(config-if-range)# switchport port-security mac-address sticky
S3(config-if-range)#exit
S3(config)#

```

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8. SSH sur les routeurs

➤ Désactiver Telnet et activer SSH sur R1, R2 et R3.

```

R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# ip domain-name reseau.local
R1(config)# crypto key generate rsa
The name for the keys will be: R1.reseau.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#username admin privilege 15 secret Cisco@123
*Mar 1 2:8:53.491: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)# login local
R1(config-line)#exit
R1(config)#

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```
R2>enable
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# ip domain-name reseau.local
R2(config)#crypto key generate rsa
The name for the keys will be: R2.reseau.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R2(config)# username admin privilege 15 secret Cisco@123
*Mar 1 2:10:53.642: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2(config)#line vty 0 4
R2(config-line)#transport input ssh
R2(config-line)#login local
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password Cisco@123
R2(config-line)#login
R2(config-line)#exit
R2(config)#ip ssh version 2
R2(config)#
```

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```
R3>enable
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip domain-name reseau.local
R3(config)#crypto key generate rsa
The name for the keys will be: R3.reseau.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#username admin privilege 15 secret Cisco@123
*Mar 1 2:14:20.796: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#line vty 0 4
R3(config-line)#transport input ssh
R3(config-line)#login local
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password Cisco@123
R3(config-line)#login
R3(config-line)#exit
R3(config)#ip ssh version 2
R3(config)#
```

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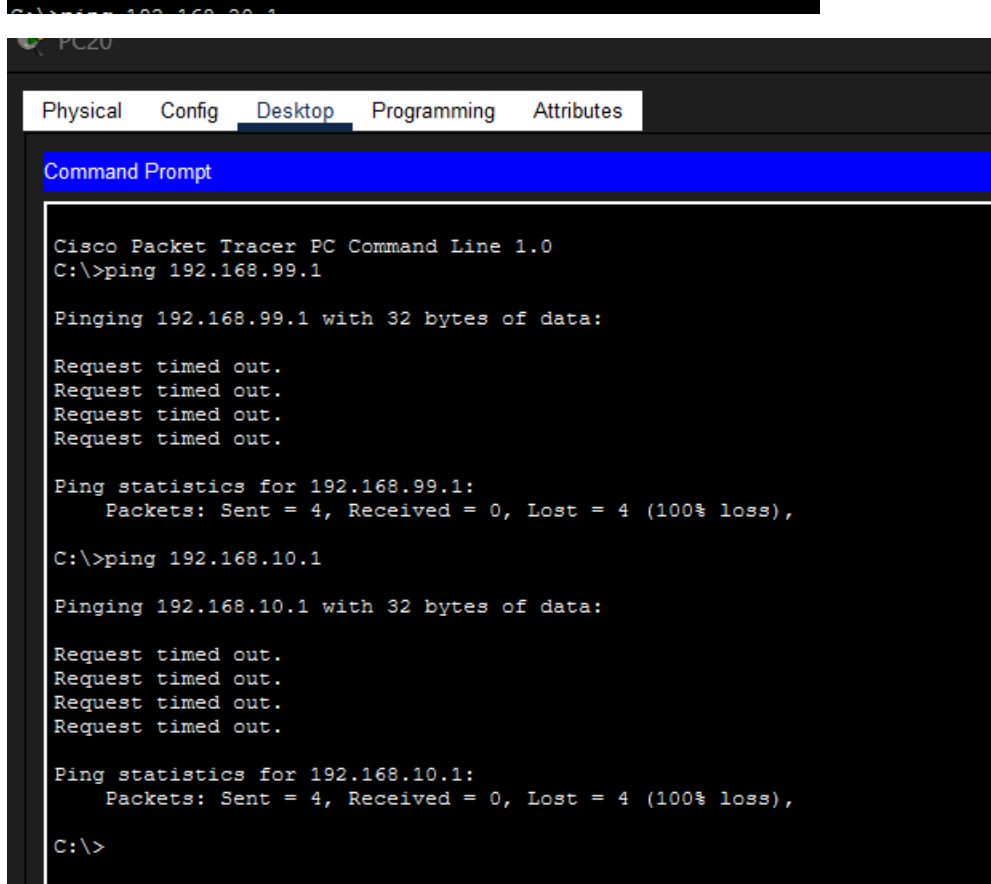
❖ Scenario de Test :

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=9ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=24ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 8ms
```



✓ Verification de l'ACL sur R3

```
User Access Verification

Password:

R3>enable
R3#show access-lists
Extended IP access list 120
 10 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
 20 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
 30 permit ip any any

R3#
```

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✓ **Vérification** ; Telnet désactivé, SSH actif

PC1

Physical Config **Desktop** Programming Attributes

Command Prompt X

```
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping192.168.20.3
Invalid Command.

C:\>clear
Invalid Command.

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>telnet 192.168.10.1
Trying 192.168.10.1 ...
% Connection timed out; remote host not responding
C:\>
```

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✓ Connexion ssh établie

```
User Access Verification

Password:

R2>enable
R2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R2#
```

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✓ Vérification sur le switch

Switch0

Physical Config CLI Attributes

Fa0/1	2	0	0	Restrict
Fa0/2	2	0	0	Restrict
Fa0/3	2	1	0	Restrict
Fa0/4	2	1	0	Restrict
Fa0/5	2	0	0	Restrict
Fa0/6	2	1	0	Restrict
Fa0/7	2	1	0	Restrict
Fa0/8	2	0	0	Restrict
Fa0/9	2	0	0	Restrict
Fa0/10	2	0	0	Restrict

```
S1#show port-security interface fa0/3
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 2
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0002.16D1.2680:10
Security Violation Count : 0

S1#
```

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Conclusion

Ce TD m'a permis de mettre en pratique les concepts essentiels de la sécurité réseau sur Cisco Packet Tracer. J'ai appris à segmenter un réseau en plusieurs VLANs pour isoler les départements, à configurer le routage inter-VLAN via des sous-interfaces, et à appliquer des règles de sécurité avancées telles que les ACL étendues, le Port Security et SSH.